

Semester II				
IT21: Java Programming				
Teaching Scheme: Theory Sessions: Total 45 Hours		Credit: 03	Examination Scheme: Internal (TH): 25 Marks External (TH): 50 Marks Total :75 Marks	
Prerequisites: Learner should know programming structures like decision flows, loops, variables, and function etc.				
Course Objectives: • To familiarize students with the concepts of OOPs. • To enable the students to understand the core principles of the Java Language and use AWT tools to produce well designed, effective applications. • Students will be able to develop server-side applications with database handling using servlets, JSP, JDBC				
Course Outcomes: On completion of the course, learners should be able to				
CO#	Cognitive Domain	Course Outcomes		
CO1	Apply	Apply the concept of Object-Oriented Programming to map and solve simple real world problem		
CO2	Apply	To design and develop robust, efficient, multithreaded and scalable Java applications using the collection framework, multithreading, and exception handling.		
CO3	Apply	To develop Web application for solving real life problem using Servlet		
CO4	Apply	To develop Web application for solving real life problem using JSP, JDBC		
CO5	Apply	To develop robust web applications using Spring MVC		
Unit No.	Contents		Weightage in %	No of Sessions
1	Basics of Java 1.1 Class and objects 1.3 Abstraction, polymorphism inheritance, and encapsulation, 1.4 Abstract Class, Interface 1.4 Garbage Collector 1.5 Lambda expression		25	10
*Mapping of Course Outcomes for Unit 1: CO1				
2	Advanced Java Concepts 2.1 Introduction to Collection Framework 2.1.1 ArrayList, Vector, Set, Map, Hashing 2.2 Multithreading		20	9

	2.2.1 Thread Life-Cycle 2.2.2 Thread Priorities 2.2.3 Synchronizing Threads 2.2.4 Inter Communication of Threads 2.3 Exception Handling 2.3.1 Types of Exception 2.3.2 Keywords 2.3.3 User defined exception		
*Mapping of Course Outcomes for Unit 2: CO2			
3	Servlets 3.1 Fundamentals of Java Servlet programming 3.2 A simple java Servlet 3.3 Servlet life cycle 3.4 Developing and Deploying Servlets 3.5 Working with cookies	20	9
*Mapping of Course Outcomes for Unit 3: CO3			
4	Java Server Pages 4.1 JSP Overview-Installation- 4.2 JSP Tags-Components of a JSP page 4.3 Expressions Script lets-Directives, JSP object, 4.4 JDBC connectivity	20	9
*Mapping of Course Outcomes for Unit 4: CO4			
5	Spring MVC 5.1 Overview of the Spring Framework 5.2 Spring MVC Annotation 5.3Spring MVC Architecture 5.4 Spring MVC Flow, 5.5 Spring Form Handling 5.2 Spring Core and Spring Boot Dependency injection and inversion of control (IoC)	15	8
*Mapping of Course Outcomes for Unit 5: CO5			
Learning Resources			
Text Books • Java Complete Reference Schildt Herbert, TMH. • Java Fundamentals (SIE), Schildt Herbert, TMH • The Complete Reference JSP, Phil Hanna, TMH • JDBC, Servlet and JSP, Black Book, Santosh Kumar K. Dremtech publication			
Reference Books • Head First Servlets and JSP, 2nd Edition by Bert Bates, Bryan Basham, Kathy Sierra • OCPJ Oracle Certified Programmer for Java Study Guide by Kathy Sierra and Bert Bates.			

- A Programmer's Guide to Java OCJP Certification (A Comprehensive Primer) by Khalid A. Mughal and Rolf W. Rasmussen.
- Java Server Programming Java Ee&(J2EE 1.7), Black Book, Wiley publications

Recommended Learning Material

- <https://docs.oracle.com/en/java/javase/index.html>
- www.nptelvideos.com
- <https://www.geeksforgeeks.org/courses/search?query=java>

Recommended Certification

- Oracle Certified Associate Java Programmer OCA
- Oracle Certified Professional Java Programmer OCP